

B.Tech. Degree I Semester Regular Examination November 2023**EE/EC/CS/IT 23-200-0104B INTRODUCTION TO ELECTRICAL ENGINEERING**

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Apply elementary principles for finding the DC response of Circuits.

CO2: Develop and solve models of basic magnetic and electromagnetic circuits.

CO3: Apply elementary principles for finding the sinusoidal steady state features of Circuits.

CO4: Familiarize with the basic engineering principles of some common electrical systems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

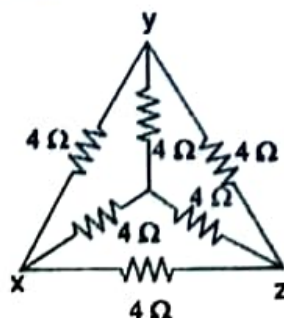
(5 × 2 = 10)

	Marks	BL	CO	PO
I. (a) Define the concept of linear and unilateral elements with necessary examples.	2	L1,L2	1	1
(b) Compare electric and magnetic circuits.	2	L1,L2	2	1
(c) Identify the resistive and reactive components of a load that consumes active power of 1.6 kW and reactive power of 1.2 kVAR when fed at 250 V.	2	L3	3	1
(d) A parallel RLC circuit with 25 Ω resistance, 25 Ω inductive reactance and 25 Ω capacitive reactance is powered by a 250 V source. Calculate the total current delivered by the source.	2	L3	3	1
(e) Briefly explain the working of RC differentiator with a neat circuit diagram.	2	L1,L2	4	1

PART B

(4 × 10 = 40)

- II. (a) Derive an expression for current in a series RC circuit, when a constant DC voltage V is applied to the circuit at $t = 0$. 5 L4 1 1
- (b) For the circuit shown below, determine the equivalent resistance between any two terminals. 5 L3 1 1

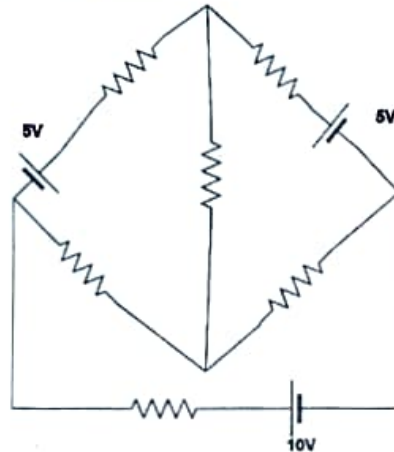


OR

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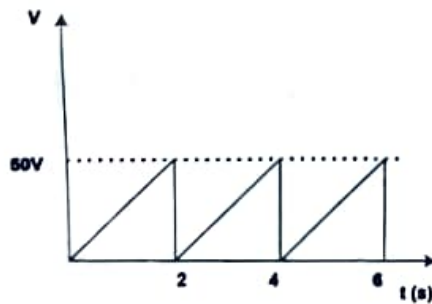
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	Marks	BL	CO	PO
III. (a) Determine the branch currents in the network shown below, when the value of each resistance is one ohm.	5	L3	1	1



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|---|---|----|---|---|
| (b) Two capacitors of capacitance $C_1 = 6 \mu\text{F}$ and $C_2 = 3 \mu\text{F}$ are connected in series across a cell of emf 18 V. Calculate: | 5 | L3 | 1 | 1 |
| (i) The equivalent capacitance | | | | |
| (ii) The potential difference across each capacitor | | | | |
| (iii) The charge on each capacitor. | | | | |

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|---|---|----|---|---|
| IV. (a) Find V_{avg} and V_{rms} of the given waveform. | 5 | L3 | 2 | 1 |
|---|---|----|---|---|

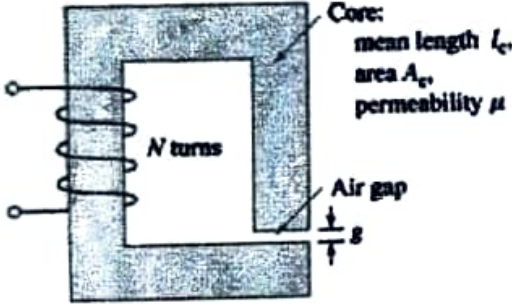


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|--|---|----|---|---|
| (b) Two coils connected in series-aiding fashion have a total inductance of 250 mH. When connected in series opposing configuration, the coils have a total inductance of 150 mH. If the inductance of first coil is three times the other, find the self inductances and the mutual inductance. | 5 | L3 | 2 | 1 |
|--|---|----|---|---|

OR

(Continued)

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		Marks	BL	CO	PO
V.	A magnetic circuit with a single air gap is shown below. The core dimensions are: Cross-sectional area $A_c = 1.8 \times 10^{-3} \text{ m}^2$ Mean core length $l_c = 0.6 \text{ m}$ Gap length $l_g = 2.3 \times 10^{-3} \text{ m}$ $N = 83$ turns  Assume that the core is of infinite permeability, μ and neglect the effects of fringing fields at the air gap and leakage flux. Calculate: (i) the reluctance of the core, S_c and that of the gap S_g (ii) the total flux ϕ and the coil inductance L for a current of $i = 1.5 \text{ A}$.	10	L3	2	1
VI.	(a) A 400 V, three phase source feeds a balanced three phase load with per phase impedance of $(32-j24) \Omega$. Compute the <u>line current, phase current, active power, reactive power and apparent power</u> if the load is (i) star connected (ii) delta connected. (b) An impedance of $(16+j14) \Omega$ and an admittance of $j0.5 \text{ S}$ are connected in parallel. Find out the effective admittance, effective impedance and power factor of the circuit. How do they share a current of 10 A?	5	L3	3	1
	OR				
VII.	(a) A 250 V, 50 Hz feeds an RLC series circuit with $R = 5 \Omega$, $L = 30 \text{ mH}$ and $C = 250 \mu\text{F}$. Calculate the current through the circuit, voltage across each element of the circuit and pf of the circuit. (b) A two element series circuit draws a current of $i(t) = 8\sqrt{2} \sin(100\pi t - 36.87^\circ) \text{ A}$ from a voltage source with instantaneous value $v(t) = 240\sqrt{2} \sin 100\pi t \text{ V}$. Determine the values of the passive elements of the circuit.	5	L3	3	1
VIII.	(a) Explain the various levels of power transmission and distribution stages with a neat single line diagram. (b) Briefly explain the functional block diagram of online UPS.	5	L1,L2	4	1
	OR				
IX.	(a) Explain the need and different methods of galvanic isolation. (b) Briefly explain the functional block diagram of BLDC fans.	5	L1,L2	4	1

Bloom's Taxonomy Levels

L1 = 14.44%, L2 = 14.44%, L3 = 65.55%, L4 = 5.55%.
